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SECTION: A

REQUIREMENTS TABLE

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically generate seeded single-elimination or double-elimination tournament brackets based on initial team rankings.	High	Pass: When valid teams, initial rankings, and a valid tournament format are provided, the system generates the corresponding seeded bracket with all participating teams assigned to valid bracket slots. Fail: If required team rankings are missing or invalid, the bracket is not generated.	Provides the core functionality for organizing tournament matches and ensures teams are correctly seeded.
FR-002	Functional	The system shall allow the Tournament Director to register teams and record their initial rankings before generating a tournament bracket.	High	Pass: When valid team details and an initial ranking are entered, the system successfully registers the team and displays it in the tournament team list. Fail: Incomplete or duplicate team information is rejected.	Accurate team registration and rankings are required for correct bracket generation.
FR-003	Functional	The system shall allow an authorized match official to record and update live scores for an ongoing match.	High	Pass: When a valid score is submitted for an ongoing match, the system updates and displays the current match score. Fail: Invalid score information is rejected and the previous valid score remains unchanged.	Live score recording is necessary for monitoring matches during the tournament.
FR-004	Functional	The system shall automatically determine the winner of a completed match and advance the winning team to the appropriate next bracket slot.	High	Pass: When a valid final match result is recorded, the system identifies the winning team and places it in the appropriate next-round bracket slot. Fail: A team is not advanced when the match result is incomplete or invalid.	Automatic winner advancement reduces manual effort and prevents errors in tournament progression.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall update tournament standings after a valid completed match result is recorded.	Medium	Pass: When a valid completed match result is recorded, the system recalculates and displays the updated tournament standings. Fail: Standings remain unchanged when an invalid match result is submitted.	Updated standings allow teams and tournament participants to track tournament progress.
NFR-001	Nonfunctional	The system shall recalculate standings, goal differentials, and tournament leaderboard updates within 200 ms of score submission.	High	Pass: During performance testing under simulated peak load, at least 95% of valid score submissions result in updated standings, goal differentials, and leaderboard information within 200 ms. Fail: More than 5% of valid score submissions exceed the 200 ms limit.	Ensures fast tournament updates during live match activity and maintains responsive system performance.
NFR-002	Nonfunctional	The system shall restrict tournament-management operations to authenticated users according to their assigned roles.	High	Pass: Authorized users can perform only the tournament-management operations permitted by their assigned roles, while unauthorized users are denied access in all security tests. Fail: An unauthorized user successfully performs a restricted tournament-management operation.	Protects tournament data and prevents unauthorized modification of brackets, scores, and standings.